Interview summary: exp_004

(Former P_04 in the focus group)

Fairness

Q: Does the implementation decision described in the statements involve an ethically relevant consideration?

*Overall, I would say yes. F1, F2, and F4 clearly involve ethically relevant considerations because they concern the use of demographic characteristics to shape development choices (e.g., to *arbitrarily* impose additional verification procedures to a specific group of users basing for example on their nationality). In a sense, I would say that choices for F1, F2 and F4 can have the potential to change the "output", understood as the execution of the code.*
As for F3, I think that it has potential ethical relevance through age-related stereotyping (which is related to the way they are represented/considered) , but this is less explicit: the wording could in fact reflect a stereotypical assumption ("older users need the guided mode"), but the comment does not itself change users' access to the guided mode. It's a bit more subtle than the others.
*In any case, I would again say that all of the statements involve an ethically relevant consideration, since they all have an ethical *impact*, regardless of how this might manifest in practice.*

Q: Do the statements primarily reflect the intended ethical aspect, rather than the others?

Yes, especially for F2, F3 and F4.
*F1 also primarily reflect the Fairness, so it's aligned with the others.. but in some way it can overlap with Privacy (that concerns also the *use* of personal data). I mean, even though gender and nationality are already available, deciding whether to use them for ranking could be intended as a decision on personal data use (a user might interpret it this way, like: "Who gave you permission to use my gender to rank recommendations?")*
*Then, it can also be that a developer might decide to use it because they believe it will improve the *accuracy* of recommendations - apparently without taking into account the influence that a data piece, such as the gender, might improperly exert on the output. What is mean is that the ability to identify ethical implications is, indeed, an "ability", and a way of observing and thinking about the consequences of choices and actions - but not everyone has it.*
I imagine that here a developer's response could therefore reflect either fairness-related or privacy-related concerns: to answer this without confusing the two things, the scenario/statement could be made clearer what is at stake from a truly ethical - and truly fairness-based - perspective.

Q: Do the alternative implementation choices in the scenarios differ meaningfully in the extent to which they reflect the ethical aspect?

Yes, especially for F2, F3 and F4: there is a meaningful contrast between applying a common procedure and introducing group-based differences etc...
For F1, what I said above goes here (the thing about the ability to identify fairness-related consequences): the alternatives differ in whether demographic attributes are used, but this alone does not establish which implementation is "fairer". A little more context would make this clearer.

Q: Could a developer's response to this scenario provide evidence about their preference regarding the ethical aspect?

Yes, with a reservation: I think that the responses should be interpreted as indicative of their fairness-based profile, because a response also depends on how the developer understands the scenario and whether they recognize its fairness-related implications.
To ensure that the responses provide a meaningful account of the developer's fairness, probably we should make sure that the way the scenarios and the consequences of the choices are interpreted is the same as that of those who designed the scenarios to elicit evidence (and of those who, in a second phase, will read the ethical profile to apply it, i.e., the LLM).
Responses are relevant as long as the lens through which they are given and interpreted remain the same.

Q: Additional comments (if any)

I don't know if respondents will see the bolded nicknames in the statements, but it seems to me that a name like "Stereotypes in code/comments" might already suggest a judgement or even a correct answer: if I were a developer, I would realize that in that scenario, my choice is being called into question in light of a possible stereotypical view I might hold.

Inclusiveness

Q: Does the implementation decision described in the statements involve an ethically relevant consideration?

Yes

Q: Do the statements primarily reflect the intended ethical aspect, rather than the others?

Yes

Q: Do the alternative implementation choices in the scenarios differ meaningfully in the extent to which they reflect the ethical aspect?

Yes, although the difference was not immediately apparent to me in I1.
As someone who is not a developer, I initially needed to reflect on which users would benefit from keyboard navigation and why this would improve accessibility (provided that this may already be clear to developers that will answer the questionnaire :))
Maybe a possible user could be a blind person that can navigate through keyboard commands rather than needing to visually locate a pointer with the mouse? If I got it right, than the implication became clearer to me.
Anyway, making the accessibility-related benefit slightly more explicit could help ensure that their answers reflect their preferences in this regard..

Q: Could a developer's response to this scenario provide evidence about their preference regarding the ethical aspect?

I would say yes.

Q: Additional comments (if any)

One difference I noticed compared with the Fairness statements is that the Inclusiveness scenarios consistently make the practical implementation trade-off explicit. Also the phrasing in the scenarios are very consistent and familiar from I1 to I4.
Then, I also noticed that the more inclusive option is described as requiring additional implementation, or testing, etc., while the less inclusive option is presented as simpler to implement.
In general , I appreciate that this provides additional context for the decision, but on the Fairness scenarios, implementation effort is not mentioned, and the fairness-related implications often need to be inferred. Inclusive scenarios are much more explicit, Fairness ones seem more subtle.
I am not suggesting that either approach is necessarily better, but I wanted to highlight this difference in how the choices are framed across the two aspects.If that's the way it's meant to be, then it makes sense :)

Privacy

Q: Does the implementation decision described in the statements involve an ethically relevant consideration?

Yes

Q: Do the statements primarily reflect the intended ethical aspect, rather than the others?

Yes

Q: Do the alternative implementation choices in the scenarios differ meaningfully in the extent to which they reflect the ethical aspect?

Overall, yes. However, the interpretation is less straightforward in P2 and P3, where using more precise data or retaining information is explicitly subject to the user's permission or consent and seems to provide an additional benefit.
A developer could choose these options to respect an informed user preference (after all, it remains a possible option that the user can opt out), and not necessarily because they place less importance on Privacy.

Q: Could a developer's response to this scenario provide evidence about their preference regarding the ethical aspect?

I find it difficult to interpret these responses as evidence of a developer's Privacy preferences, particularly in P2 and P3, where the decision is explicitly mediated by the user's permission or consent, as mentioned in the previous response. Beyond the stated legal and organizational constraints, I see this primarily as the user's choice rather than the developer's.
Putting myself in a developer's position: even if I strongly value privacy, why should I prevent a user from voluntarily sharing their precise location to receive more accurate recommendations? Why should I decide in advance not to ask for it at all?
Am I supposed to have concerns, e.g., about whether these data could be used unlawfully within the company/organization? Could this be a reason why I might decide not to request certain data at all?
However, similar concerns would bring us back to regulation and transparency towards the user, which seem to fall outside the stated scope of these scenarios..
I mean, allowing that choice does not necessarily mean placing less importance on privacy..

Q: Additional comments (if any)

No additional comments

Transparency and Explainability

Q: Does the implementation decision described in the statements involve an ethically relevant consideration?

Overall, I would say yes. TE3, TE4 and TE5 have more direct ethical relevance because they concern the information users receive about data requests, validation outcomes, and the consequences of their actions.
TE1 and TE2 also have potential ethical relevance through making implementation decisions easier to understand and review, although this is less explicit: as written, they may primarily appear to concern code readability and documentation rather than an ethical consideration. What I mean is that the "target" (who benefits or loses out from certain decisions) is clearer in TE3, TE4 and TE5, i.e., the user - and less clear in TE1 and TE2. By consequence, also the relevance of ethical consideration appears clearer when the scenario is more explicit in this sense.

Q: Do the statements primarily reflect the intended ethical aspect, rather than the others?

Yes

Q: Do the alternative implementation choices in the scenarios differ meaningfully in the extent to which they reflect the ethical aspect?

*Yes, even If the contrast is somewhat less sharp in TE1: named variables may improve readability (i.e., explainability of the code), but a compact conditional expression may not necessarily be difficult to understand.. so I would say that here choices do not differ so *meaningfully*.*

Q: Could a developer's response to this scenario provide evidence about their preference regarding the ethical aspect?

Yes

Q: Additional comments (if any)

Naming the scenario with the term "Meaningful" (T2) conveys a judgment and tends to steer one toward a choice that seems more correct.

Traceability

Q: Do the statements primarily reflect the intended ethical aspect, rather than the others?

Yes

Q: Do the alternative implementation choices in the scenarios differ meaningfully in the extent to which they reflect the ethical aspect?

Yes

Q: Could a developer's response to this scenario provide evidence about their preference regarding the ethical aspect?

*I would say yes, that the responses could provide evidence about a developer's preference for traceability. However, interpreting all of this as an *ethical* preference is less straightforward, because in the scenarios, the ethical load is unclear - unlike, for example, in the scenarios on transparency or privacy, or even in what I imagine the scenarios on resource consumption to be like.*
A developer might choose better tracing for practical reasons (I was thinking for example about reasons such as debugging, or to create reports, documentations, etc...) rather than for its ethical implications. Probably the ethical relevance of these choices could be made clearer.
This also relates to my earlier observation about TR2, where the ethical significance of the action being traced is less explicit, and so is the evidence about the developer's preference about this.

Q: Additional comments (if any)

No further comments

Resource Consumption

Q: Does the implementation decision described in the statements involve an ethically relevant consideration?

Yes.. I recognize that the implementation choices can affect resource consumption and therefore have ethical relevance through their potential implications for energy use and sustainability - and probably a developer can recognize this better than me.
However, this connection is particularly implicit here for RC scenarios, which may appear to concern technical efficiency (the consequence at stake seem to be a technical one in every

statement). The significance of the ethical impact would also depend on the scale of the resource consumption involved, provided that it is correctly identified by those who read the scenarios.

Q: Do the statements primarily reflect the intended ethical aspect, rather than the others?

Yes

Q: Do the alternative implementation choices in the scenarios differ meaningfully in the extent to which they reflect the ethical aspect?

Yes

Q: Could a developer's response to this scenario provide evidence about their preference regarding the ethical aspect?

Again, assuming they have a clear understanding of the impact of each individual scenario in terms of resource and energy consumption - and that they remember that this is the deciding factor in their choice, and not other reasons - then yes.
Additional comments (if any)
Naming the scenario with the term "Redundant" or "Unnecessary" or "Disproportionate" conveys a judgment and tends to steer one toward a choice that seems more correct.

Final Questions

Q: Do you think that the proposed five-point response scale is appropriate for expressing the developer's preference regarding the described implementation choice?

Yes, the five-point response scale seems appropriate.
I always found it difficult to understand what factors might distinguish a 1 from a 2 - or a 4 from a 5 - and how the responses might then be interpreted differently once aggregated with the others. But overall, I consider it appropriate for this kind of scenarios that can lead to more nuanced reflections.

Q: Do you think that averaging the statements scores is an appropriate aggregation strategy?

Probably, but I don't have a clear idea on this.

Q: Should all statements contribute equally to the aspect score?

As I mentioned during the focus group, I do not think that all statements should necessarily contribute equally to the aspect score.
In my view, the scenarios differ in their ethical significance, and this also changes the weight of the decision the developer is asked to make. Where the ethical stakes are limited, or the difference between the alternatives is not particularly meaningful from an ethical perspective, I would not give the response the same weight as a response to a scenario in which the intended ethical aspect is more fully represented and the consequences of the choice are more substantial.
I cannot immediately suggest a precise criterion for determining these weights, but some differences seem clear to me. For example, responses to F2 (Differences in validation procedures) or I3 (Gender identities) may tell us more, ethically speaking, than a response to TE1 (Self-explanatory code structure). Even within the same aspect, I would consider TE3 (Data request explanation) to carry greater ethical weight than TE1.
My point is not simply that some items should have a greater numerical influence on the final profile, but also that the weighting should acknowledge differences in the ethical significance of the choices themselves, including the sensitivity required of the developer to recognize their potential impacts and consequences for single users, communities, environments, and the public in general.

Authors' summary

- Concerning the statement F3, the interviewed reports that the comment written by the developer remains into the code and do not manifest in its behavior.
  Authors' response: this is in line with the idea that ethical manifestations in the written code also concern the code as a textual artifacts, and its impact concern the developer-to-deveoper relationship.

- F1 has also a privacy concern and the additional information could enhance the results accuracy. In general, the ethical implications are implicit and depend on the developer's ability to notice them while deciding over an implementation choice.
  Authors' response: data are already available in the system, and were already collected. We can make it more clear. The statements leave the ethical implications implicit exactly for the purpose of stressing the developer's sensitivity to the topics, and to avoid biases that could arise when ethical considerations are made explicit.

- For I1, the reasons for keyboard navigation should be explained

- Asking the context to be used for all the scenarios is a good decision

- It should be highlighted that the implementation effort is the same (sometimes it is explicit, sometimes no)
  Authors' response: we can make it explicit in the questionnaire instructions

- In P2, asking the user's consent would simplify the privacy issue: the decision is delegated to the user

- Words like "redundand", "unnecessary", or "disproportionate" should be avoided to reduce bias

- The 5-point scale is ok, since answers provide more guidance than a simple 1 to 5 score. Qualitative options may be more informative for the respondents.

- While into the same category there are no inconsistencies, categories have different ethical relevance.